WEEK 9: Sales and Customer Analytics System

Ceramic Creations: From Clay to Art requires a Sales and Customer Analytics System to analyze customer purchases, sales performance, and product trends. The system helps identify top-selling ceramic products, frequent customers, and revenue patterns.

Feature	Description
SUM()	Calculates the total sales revenue generated from ceramic products.
COUNT()	Counts the total number of orders, customers, and products sold.
AVG()	Determines the average order value and customer spending.
MAX()	Identifies the highest-selling product and maximum sales amount.
MIN()	Finds the lowest-selling product and minimum sales amount.

1.Total number of orders placed

```
SQL> SELECT COUNT(*) AS TOTAL FROM ORDERS;

     TOTAL
----------
        11
```

2.Total revenue generated from sales

```
SQL> SELECT SUM(TOTAL_AMOUNT) AS REVENUE FROM ORDERS;

   REVENUE
----------
      6305
```

3.Average order value

```
SQL> SELECT AVG(TOTAL_AMOUNT) AS AVERAGE FROM ORDERS GROUP BY ORDER_ID;

   AVERAGE
----------
       550
       250
       600
       150
       150
       125
      1000
      1000
      1500
       580
       400

11 rows selected.
```

```
SQL> SELECT AVG(TOTAL_AMOUNT) AS AVERAGE FROM ORDERS;

   AVERAGE
----------
573.181818
```

4.Highest and lowest order amount

```
SQL> SELECT MAX(TOTAL_AMOUNT) AS MAXIMUN, MIN(TOTAL_AMOUNT) AS MINIMUM FROM ORDERS;

   MAXIMUN    MINIMUM
---------- ----------
      1500        125
```

5.Total number of orders placed by each customer

```
SQL> SELECT CUSTOMER_ID,COUNT(*) AS TOTAL FROM ORDERS GROUP BY CUSTOMER_ID;

CUSTOMER_ID      TOTAL
----------- ----------
        101          1
        102          1
        103          1
        104          1
        105          2
        106          1
        107          1
        108          1
        109          1
        110          1

10 rows selected.
```

6.Total amount spent by each customer

```
SQL> SELECT CUSTOMER_ID,SUM(TOTAL_AMOUNT) AS TOTAL FROM ORDERS GROUP BY CUST
OMER_ID;

CUSTOMER_ID      TOTAL
----------- ----------
        101        550
        102        250
        103        600
        104        150
        105        550
        106        125
        107       1000
        108       1000
        109       1500
        110        580

10 rows selected.
```

7.Products generating highest revenue

```
SQL> SELECT P.PRODUCT_ID,P.PRODUCT_NAME,SUM(OD.QUANTITY) AS QUANTITY,SUM(TOTAL_PRICE) AS TOTAL_REVENUE
  2  FROM PRODUCT P JOIN ORDER_DETAILS OD
  3  ON P.PRODUCT_ID=OD.PRODUCT_ID
  4  GROUP BY P.PRODUCT_ID,P.PRODUCT_NAME
  5  ORDER BY TOTAL_REVENUE DESC;

PRODUCT_ID PRODUCT_NAME      QUANTITY TOTAL_REVENUE
---------- -------------- ---------- -------------
P00009     Keychain               10          1500
P00008     Spoons                  4          1000
P00007     Chip_Dip boul           2          1000
P00002     Coffee Mug              4           600
P00010     Engraved Plates         2           580
P00004     Flower_Vase             2           550
P00011     Book Mark               4           400
P00006     Ceramic Plate           2           300
P00001     Ceramic bowl            2           250
P00003     Plant Pot               1           150
P00005     Wall hanging            1           125

11 rows selected.
```

8.Total sales generated by each category

```
SQL> SELECT C.CATEGORY_NAME,SUM(OD.TOTAL_PRICE)AS TOTAL_REVENUE
  2  FROM CATEGORY C JOIN PRODUCT P
  3  ON C.CATEGORY_ID=P.CATEGORY_ID
  4  JOIN ORDER_DETAILS OD
  5  ON P.PRODUCT_ID=OD.PRODUCT_ID
  6  GROUP BY C.CATEGORY_ID,CATEGORY_NAME;

CATEGORY_NAME        TOTAL_REVENUE
-------------------- -------------
Home_Decor                     425
Kitchenware                   2850
Garden_Outdoor                 700
Custom Creations              1500
Personalized Gift              580
Unique Creations               400

6 rows selected.
```

Report 1 : Sales Performance Report

```
SQL> SELECT COUNT(*) AS TOTAL, SUM(TOTAL_AMOUNT) AS AMOUNT,AVG(TOTAL_AMOUNT)
 FROM ORDERS;

     TOTAL     AMOUNT AVG(TOTAL_AMOUNT)
---------- ---------- ------------------
        11       6305        573.181818
```

Report 2 : Customer Analytics Report

```
SQL> SELECT C.CUSTOMER_NAME,COUNT(O.ORDER_ID) AS NUMBER_OF_ORDERS,COUNT(O.ORDER_ID) AS FREQUECCE_PURCHASE
  2  FROM CUSTOMER C LEFT JOIN ORDERS O
  3  ON C.CUSTOMER_ID=O.CUSTOMER_ID
  4  GROUP BY C.CUSTOMER_NAME;

CUSTOMER_NAME            NUMBER_OF_ORDERS FREQUECCE_PURCHASE
------------------------ ---------------- ------------------
Sweety                                  1                  1
Arun                                    1                  1
Priya                                   1                  1
Kumar                                   1                  1
Meena                                   2                  2
Ravi                                    1                  1
Ananya                                  1                  1
Snaha                                   1                  1
Rahul Verma                             1                  1
Karthik                                 1                  1
Vihan                                   0                  0

11 rows selected.
```

Report 3 : Product Performance Report

```
SQL> SELECT P.PRODUCT_NAME,OD.QUANTITY,OD.TOTAL_PRICE
  2  FROM PRODUCT P JOIN ORDER_DETAILS OD
  3  ON P.PRODUCT_ID=OD.PRODUCT_ID;

PRODUCT_NAME      QUANTITY TOTAL_PRICE
--------------- ---------- -----------
Coffee Mug                4         600
Flower_Vase               2         550
Wall hanging              1         125
Ceramic bowl              2         250
Plant Pot                 1         150
Ceramic Plate             2         300
Chip_Dip boul             2        1000
Spoons                    4        1000
Keychain                 10        1500
Engraved Plates           2         580
Book Mark                 4         400

11 rows selected.
```

Report 4 : Category Analysis Report

```
SQL> SELECT C.CATEGORY_NAME,SUM(OD.QUANTITY) AS QUANTITY,SUM(OD.TOTAL_PRICE)
AS TOTAL_REVENUE
  2  FROM CATEGORY C JOIN PRODUCT P
  3  ON C.CATEGORY_ID=P.CATEGORY_ID
  4  JOIN ORDER_DETAILS OD
  5  ON P.PRODUCT_ID=OD.PRODUCT_ID
  6  GROUP BY C.CATEGORY_ID,C.CATEGORY_NAME;

CATEGORY_NAME          QUANTITY TOTAL_REVENUE
-------------------- ---------- -------------
Home_Decor                    3           425
Kitchenware                  12          2850
Garden_Outdoor                3           700
Custom Creations             10          1500
Personalized Gift             2           580
Unique Creations              4           400

6 rows selected.
```

Conclusion

The **Sales and Customer Analytics System** helps *Ceramic Creations: From Clay to Art* understand customer purchasing behavior and sales performance. It provides valuable insights into product demand, customer preferences, and business trends. The system supports effective decision-making, improves customer satisfaction, and contributes to overall business growth.